



Introduction to Open Systems

LECTURE 1 COM 225 LECTURE NOTES

BY, DR. GILBERT MUGENI, PH.D

Learning Outcomes

Define Open Source Systems and related concepts.

Differentiate between open source and proprietary software.

Explain the historical development of the open source movement.

Identify key milestones in open source history.

Discuss advantages and limitations of open source systems.

What is Open Source?

- Software whose source code is publicly available
- Users can view, modify, and distribute the code
- Encourages collaboration and transparency
- Often developed by global communities
- Supported by open licensing models

Characteristics of Open Source Software

- Source code is accessible to everyone
- Software can be modified and improved
- Redistribution is allowed
- Community-driven development
- Transparency in design and development

Evolution of Open Source Systems

- Early computing emphasized sharing of code
- Commercial software models emerged in the 1970s–80s
- Proprietary software restricted access to source code
- Developers responded with open collaboration models
- Open source movement gained global momentum

Early Roots of Open Source

- Collaboration among early computer scientists
- Software shared freely in academic institutions
- Code exchange was common in research communities
- Lack of strict intellectual property restrictions
- Foundation for modern open source practices

Rise of free software movement

- Initiated by software freedom advocates
- Focused on user rights to run and modify software
- Encouraged collaborative software development
- Emphasized ethical and social aspects of software
- Influenced modern open source philosophy

Open Source software movement

- Term “Open Source” introduced in the late 1990s
- Promoted collaborative and transparent development
- Focused on practical benefits of open software
- Encouraged business adoption of open models
- Led to growth of open source projects worldwide

Open Source Initiative

- Organization promoting open source principles
- Defines standards for open source licenses
- Advocates for open development practices
- Encourages adoption by industry and governments
- Supports global open source communities

OSS Terminology

FOSS (Free and Open Source Software)

FLOSS (Free/Libre Open Source Software)

OSS (Open Source Software)

Source code

Open source licenses

Understanding FOSS

- Stands for Free and Open Source Software
- Emphasizes freedom to use and modify software
- Combines free software and open source concepts
- Encourages collaborative development
- Widely used in academic and research contexts

Understanding FLOSS

- Means Free/Libre Open Source Software
- “Libre” highlights freedom rather than price
- Emphasizes user rights and openness
- Promotes transparency and collaboration
- Used in global open source communities

Examples of OSS Systems

- Operating systems such as Linux
- Web servers such as Apache
- Databases such as MySQL and PostgreSQL
- Programming languages such as Python
- Development tools such as Git

Benefits of OSS

- Often free to use and distribute
- Reduces software acquisition costs
- Lower licensing fees
- Suitable for educational institutions
- Enables cost-effective innovation

Challenges of OSS

- Limited official technical support in some cases
- Compatibility issues with proprietary software
- Requires technical expertise for customization
- Risk of poorly maintained projects
- Security risks if updates are ignored

OSS in Modern Organisations

- Widely used in cloud computing platforms
- Supports digital transformation initiatives
- Encourages innovation in software development
- Adopted by governments and enterprises
- Drives collaborative technology ecosystems

Summary

- Open source software allows access to source code
- Originated from collaborative software culture
- Key terms include FOSS, FLOSS, and OSS
- Provides flexibility, cost savings, and innovation
- Also presents challenges requiring proper management